

TRADEPASS ACADEMY

309A COMPANION SERIES · VOLUME 3

30-Day Study Planner

A day-by-day route through the Construction Electrician 309A exam: daily checklists, weekly reviews, mock exam tracking and weak-area repair.

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- Thirty structured study days with daily checklists
 - Four weekly review pages that force course correction
 - Mock exam tracker with topic-by-topic scoring
 - A weak-area tracker that turns misses into marks
 - Notes pages for code references and reminders

Companion resource to: Red Seal Construction Electrician 309A Exam Prep, 2026–2027 Edition

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How this planner works

Thirty days is enough to prepare properly for the 309A examination if the days are structured, and this planner is that structure. Each day pairs with your TradePass Academy study guide: roughly sixty to ninety minutes on weekdays, longer sessions where a mock exam is scheduled.

- **Book your exam first.** A date on the calendar changes how you study. Count back thirty days and begin.
- **Tick boxes as you finish tasks, not before.** The planner only tells the truth if you do.
- **Rate your confidence every day.** A row of “Low” ratings on one topic is your weak-area tracker filling itself in.
- **Never skip the weekly review page.** It is the mechanism that redirects your remaining days to where the marks are.
- **Missed a day?** Continue where you left off rather than compressing two days into one. The sequence matters more than the calendar.

EXAM TIP

The three mock exams on Days 8, 18 and 26 are the backbone of this plan. Protect those sessions: full length, timed, one sitting, no notes and no phone.

The 30-day map

Day	Focus	Day	Focus
1	Orientation and diagnostic setup	16	Motor protection and sizing
2	Safety and safe work practice	17	Motor control circuits
3	DC circuit theory	18	MOCK EXAM 2 + scoring
4	Series and parallel calculation drill	19	Mock 2 corrections
5	AC theory I: waveforms and reactance	20	Transformers I: theory and ratios
6	AC theory II: power and power factor	21	Transformers II: connections and protection
7	Three-phase systems	22	Services, distribution and demand
8	MOCK EXAM 1 (diagnostic) + scoring	23	Lighting and emergency systems
9	Mock 1 corrections and weak-area list	24	Fire alarm, low voltage and communications
10	Conductors and ampacity	25	Renewables and special installations
11	Wiring methods and cables	26	MOCK EXAM 3 + scoring
12	Raceways, boxes and fill	27	Mock 3 corrections
13	Grounding and bonding I	28	Weak-area repair day
14	Grounding and bonding II	29	Formula index and facts drill
15	Overcurrent protection	30	Light review and rest

Week 1 — Foundations (Days 1–7)

Day 1 — Orientation and diagnostic setup

- ☐ Read the exam breakdown for Construction Electrician at red-seal.ca and note the weighting of each major work activity.
- ☐ Skim your study guide's table of contents and mark chapters you expect to be weakest.
- ☐ Set your study space, times and the dates for the three mock exams (Days 8, 18, 26).

Self-check before closing the book: Can you name the major work activities of your trade and their rough exam weighting?

Confidence today: ☐ Low ☐ Getting there ☐ Solid **Time studied:** _____ min

Day 2 — Safety and safe work practice

- ☐ Study the safety chapter: lockout, test-verify-test, arc flash, PPE selection.
- ☐ Answer every end-of-chapter question; write corrections for any miss.
- ☐ Review facts 1–10 in the Last-Minute Exam Review.

Self-check before closing the book: What is the correct first action in an incident scenario, and what current range trips a Class A GFCI?

Confidence today: ☐ Low ☐ Getting there ☐ Solid **Time studied:** _____ min

Day 3 — DC circuit theory

- ☐ Work the Ohm's law and resistance chapter, including conductor resistance.
- ☐ Reproduce the Ohm's law wheel from memory.
- ☐ Complete 15 practice questions on series, parallel and combination circuits.

Self-check before closing the book: For three parallel resistors, can you find total resistance without looking up the formula?

Confidence today: ☐ Low ☐ Getting there ☐ Solid **Time studied:** _____ min

Day 4 — Series and parallel calculation drill

- ☐ Timed drill: 20 circuit calculations at one minute each.
- ☐ Rework every miss on paper, writing the step where the error happened.
- ☐ Review Kirchhoff's laws and meter connections.

Self-check before closing the book: What voltage appears across an open in a series circuit?

Confidence today: ☐ Low ☐ Getting there ☐ Solid **Time studied:** _____ min

Day 5 — AC theory I: waveforms and reactance

- ☐ Study RMS and peak values, frequency and period, XL and XC.
- ☐ Memorize ELI the ICE man and use it on ten lead/lag questions.
- ☐ Complete the chapter questions.

Self-check before closing the book: Which way does XC move when frequency rises, and which way does XL move?

Confidence today: ☐ Low ☐ Getting there ☐ Solid **Time studied:** _____ min

Day 6 — AC theory II: power and power factor

- ☐ Study true, reactive and apparent power and the power triangle.
- ☐ Work five problems converting between kW, kVA and PF.
- ☐ Note why capacitors correct a lagging power factor.

Self-check before closing the book: A load reads 10 kVA at 0.8 PF: what is the true power, without paper?

Confidence today: ☐ Low ☐ Getting there ☐ Solid **Time studied:** _____ min

Day 7 — Three-phase systems

- ☐ Study wye and delta relationships and the $\sqrt{3}$ factor.
- ☐ Sketch both connections from memory, labelling line and phase quantities.
- ☐ Work ten three-phase current and power calculations.

Self-check before closing the book: On a 120/208 V wye system, where does the 120 V come from and where does the 208 V come from?

Confidence today: ☐ Low ☐ Getting there ☐ Solid **Time studied:** _____ min

Week 1 review

Complete this page before starting the next week. Honesty here saves days later.

Which topic this week had the lowest confidence ratings, and what exactly makes it difficult?

Which calculation errors repeated across the week?

What will you do differently in Week 2 study sessions?

Week 2 — Diagnostic and installation (Days 8–14)

Day 8 — MOCK EXAM 1 (diagnostic)

- ☐ Sit a full-length practice test from your study guide: timed, one sitting, closed book.
- ☐ Score it immediately and record the result in the Mock Exam Tracker.
- ☐ Do not review corrections tonight; record and rest.

Self-check before closing the book: Did you finish inside the time limit, and if not, by how much?

Confidence today: ☐ Low ☐ Getting there ☐ Solid **Time studied:** _____ min

Day 9 — Mock 1 corrections and weak-area list

- ☐ Review every incorrect answer; read the full explanation, then re-answer from scratch.
- ☐ Enter each miss into the Weak Area Tracker with its topic.
- ☐ Identify your three lowest-scoring topics; these get priority in Weeks 2–3.

Self-check before closing the book: Can you explain, for three of your misses, why the right answer is right?

Confidence today: ☐ Low ☐ Getting there ☐ Solid **Time studied:** _____ min

Day 10 — Conductors and ampacity

- ☐ Study insulation ratings, ambient correction, grouping derating and termination limits.
- ☐ Work eight ampacity questions involving at least one derating step each.
- ☐ Memorize the small-conductor overcurrent limits (15/20/30 A).

Self-check before closing the book: What are the four checks between a table ampacity and a final conductor choice?

Confidence today: ☐ Low ☐ Getting there ☐ Solid **Time studied:** _____ min

Day 11 — Wiring methods and cables

- ☐ Study cable types, permitted uses, support and protection requirements.
- ☐ Review clearance and support numbers (stud clearance, support intervals).
- ☐ Complete the chapter questions.

Self-check before closing the book: Where is nonmetallic sheathed cable not permitted?

Confidence today: ☐ Low ☐ Getting there ☐ Solid **Time studied:** _____ min

Day 12 — Raceways, boxes and fill

- ☐ Study conduit types, the 40 percent fill rule, and box fill counting.
- ☐ Work five box-fill counts including devices and bonds.
- ☐ Note expansion-fitting and support requirements.

Self-check before closing the book: In a box-fill count, how much does a duplex receptacle count for?

Confidence today: ☐ Low ☐ Getting there ☐ Solid **Time studied:** _____ min

Day 13 — Grounding and bonding I

- ☐ Study system grounding, the main bonding jumper and electrode types.
- ☐ Write the grounding-versus-bonding distinction in two sentences of your own.

- ☐ Review facts 51–60 in the Last-Minute Exam Review.

Self-check before closing the book: Where is the one place neutral and bond connect, and why only there?

Confidence today: ☐ Low ☐ Getting there ☐ Solid **Time studied:** _____ min

Day 14 — Grounding and bonding II

- ☐ Study bonding conductor sizing, sub-panel isolation and fault-path impedance.
- ☐ Work eight questions on bonding sizes and sub-panel wiring.
- ☐ Sketch a service and sub-panel showing neutral and bond routing.

Self-check before closing the book: In a sub-panel, which bar is isolated and which is bonded to the enclosure?

Confidence today: ☐ Low ☐ Getting there ☐ Solid **Time studied:** _____ min

Week 2 review

Complete this page before starting the next week. Honesty here saves days later.

Compare your weak-area list from Day 9 with this week's confidence ratings: what improved, what did not?

Which code-lookup skills are still slow?

What is your plan for the three priority topics before Mock 2?

Week 3 — Machines and power (Days 15–21)

Day 15 — Overcurrent protection

- ☐ Study fuse and breaker types, interrupting ratings and standard sizes.
- ☐ Work the continuous-load (80 percent) rule in five questions.
- ☐ Note where protection is sized to conductors versus special rules.

Self-check before closing the book: What must the interrupting rating of a device meet or exceed?

Confidence today: ☐ Low ☐ Getting there ☐ Solid **Time studied:** _____ min

Day 16 — Motor protection and sizing

- ☐ Study the four-layer motor protection stack and which number sizes each layer.
- ☐ Work six motor questions: conductors at 125 percent, overloads from nameplate.
- ☐ Memorize the synchronous speed series (3600/1800/1200/900).

Self-check before closing the book: Which motor number sizes the overloads, and which sizes the conductors?

Confidence today: ☐ Low ☐ Getting there ☐ Solid **Time studied:** _____ min

Day 17 — Motor control circuits

- ☐ Study two-wire and three-wire control, seal-in contacts and interlocks.
- ☐ Trace three ladder diagrams and predict behaviour after a power failure.
- ☐ Review stops-in-series, starts-in-parallel.

Self-check before closing the book: After an outage, which control scheme restarts by itself and which stays off?

Confidence today: ☐ Low ☐ Getting there ☐ Solid **Time studied:** _____ min

Day 18 — MOCK EXAM 2

- ☐ Full-length, timed, one sitting, closed book.
- ☐ Score immediately and record in the Mock Exam Tracker beside Mock 1.
- ☐ Note time remaining (or overrun) for pacing comparison.

Self-check before closing the book: Which topics improved since Mock 1, by the numbers?

Confidence today: ☐ Low ☐ Getting there ☐ Solid **Time studied:** _____ min

Day 19 — Mock 2 corrections

- ☐ Full correction cycle: re-answer every miss from scratch after reading explanations.
- ☐ Update the Weak Area Tracker; mark repaired items as resolved.
- ☐ Reset priorities: the lowest topic now owns Days 28–29 slots.

Self-check before closing the book: Is any topic still missing more than one question in four?

Confidence today: ☐ Low ☐ Getting there ☐ Solid **Time studied:** _____ min

Day 20 — Transformers I: theory and ratios

- ☐ Study turns, voltage and current ratios and kVA ratings.
- ☐ Work eight ratio and full-load-current calculations.
- ☐ Memorize: 45 kVA at 208 V three-phase → 125 A.

Self-check before closing the book: If voltage steps down 5:1, what happens to current, exactly?

Confidence today: ☐ Low ☐ Getting there ☐ Solid **Time studied:** _____ min

Day 21 — Transformers II: connections and protection

- ☐ Study delta-wye connections, polarity, protection percentages and the CT hazard.
- ☐ Sketch a 600 V delta to 120/208 V wye transformation from memory.
- ☐ Complete the chapter questions.

Self-check before closing the book: Why must a CT secondary never be opened under load?

Confidence today: ☐ Low ☐ Getting there ☐ Solid **Time studied:** _____ min

Week 3 review

Complete this page before starting the next week. Honesty here saves days later.

What did Mock 2 prove about your pacing, and what will you change for Mock 3?

Which motor or transformer detail still requires the book?

List every topic you have not yet studied; they are scheduled in Week 4, confirm none are missing.

Week 4 — Systems and the final approach (Days 22–30)

Day 22 — Services, distribution and demand

- ☐ Study service equipment, working clearances, demand factors and metering.
- ☐ Work a basic demand calculation end to end.
- ☐ Review facts 61–70 in the Last-Minute Exam Review.

Self-check before closing the book: Why is calculated load smaller than connected load?

Confidence today: ☐ Low ☐ Getting there ☐ Solid **Time studied:** _____ min

Day 23 — Lighting and emergency systems

- ☐ Study luminaire types, LED retrofits, and emergency/exit unit equipment.
- ☐ Note connection-ahead-of-switching for emergency lighting.
- ☐ Complete the chapter questions.

Self-check before closing the book: What makes “ballast bypass” more than a lamp swap?

Confidence today: ☐ Low ☐ Getting there ☐ Solid **Time studied:** _____ min

Day 24 — Fire alarm, low voltage and communications

- ☐ Study Class A versus Class B fire-alarm wiring and supervision.
- ☐ Study Class 2 circuit rules and separation from line voltage.
- ☐ Review data/communications separation practices.

Self-check before closing the book: What failure does Class A wiring survive that Class B does not?

Confidence today: ☐ Low ☐ Getting there ☐ Solid **Time studied:** _____ min

Day 25 — Renewables and special installations

- ☐ Study photovoltaic basics: energized-when-lit conductors, disconnects, labelling.
- ☐ Review any special-location chapters in your guide (wet, hazardous as applicable).
- ☐ Complete the chapter questions.

Self-check before closing the book: When are PV array conductors considered energized?

Confidence today: ☐ Low ☐ Getting there ☐ Solid **Time studied:** _____ min

Day 26 — MOCK EXAM 3

- ☐ Final full-length, timed mock under strict exam conditions.
- ☐ Score and record beside Mocks 1 and 2; the trend line matters more than the number.
- ☐ Celebrate the improvement; then rest.

Self-check before closing the book: Are you finishing with time to review flagged questions?

Confidence today: ☐ Low ☐ Getting there ☐ Solid **Time studied:** _____ min

Day 27 — Mock 3 corrections

- ☐ Correct every miss with the full re-answer cycle.
- ☐ Close out the Weak Area Tracker: every line resolved or scheduled for Day 28.
- ☐ Write your personal “top five reminders” card for exam morning.

Self-check before closing the book: What five reminders go on your card?

Confidence today: ☐ Low ☐ Getting there ☐ Solid Time studied: _____ min

Day 28 — Weak-area repair day

- ☐ Spend the whole session on the single lowest topic remaining.
- ☐ Teach the topic out loud to an empty room; explanation exposes gaps recall hides.
- ☐ Finish with ten questions on that topic only.

Self-check before closing the book: Could you now explain this topic to a first-year apprentice?

Confidence today: ☐ Low ☐ Getting there ☐ Solid Time studied: _____ min

Day 29 — Formula index and facts drill

- ☐ Rewrite the Volume 1 formula index from memory; verify and re-drill gaps.
- ☐ Read all 100 facts in Volume 2 once, marking any that still surprise.
- ☐ Run the quick-calculation shortcuts out loud.

Self-check before closing the book: Can you rebuild the formula index with no misses?

Confidence today: ☐ Low ☐ Getting there ☐ Solid Time studied: _____ min

Day 30 — Light review and rest

- ☐ Morning only: memory tricks and your top-five card.
- ☐ Prepare ID, confirmation, calculator, route and clothes.
- ☐ No study after mid-day. Sleep is the last task on this plan.

Self-check before closing the book: Is everything for tomorrow morning already by the door?

Confidence today: ☐ Low ☐ Getting there ☐ Solid Time studied: _____ min

Week 4 review

Complete this page before starting the next week. Honesty here saves days later.

Across three mocks, what was your score trend by topic?

What pacing plan will you follow in the real sitting?

What is your plan for the first five minutes of the exam?

Mock exam tracker

Record each full-length test here. Score by topic so the trend, not just the total, is visible.

Topic area	Mock 1 (Day 8)	Mock 2 (Day 18)	Mock 3 (Day 26)
Safety and fundamentals	___ / ___	___ / ___	___ / ___
Circuit and AC theory	___ / ___	___ / ___	___ / ___
Conductors and wiring methods	___ / ___	___ / ___	___ / ___
Grounding and bonding	___ / ___	___ / ___	___ / ___
Protection and distribution	___ / ___	___ / ___	___ / ___
Motors and controls	___ / ___	___ / ___	___ / ___
Transformers	___ / ___	___ / ___	___ / ___
Systems and special installations	___ / ___	___ / ___	___ / ___
TOTAL	___ / ___	___ / ___	___ / ___
Time used / allowed	___ / ___	___ / ___	___ / ___

Weak area tracker

Every question you miss earns a row. A row is closed only when you can answer a fresh question on the topic correctly and explain why.

Topic / question missed	Why I missed it	Repair action	Resolved
			☐
			☐
			☐
			☐
			☐
			☐
			☐
			☐
			☐
			☐
			☐
			☐
			☐
			☐
			☐

Notes

Code references, values to memorize, questions for your instructor.

[illegible]

Notes (continued)

[illegible]
